Satya Raj Awasthi

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Research Interests

Software engineering, AI for software engineering, and programming systems and program analysis, especially at their intersections; also databases and distributed systems. I am particularly interested in building software systems that are reliable, scalable, and less error-prone, and in how agentic AI pipelines can be used in practice across every stage of the software development lifecycle (development, debugging, testing, and deployment), not just for code generation.

Education

Pokhara University (NCIT, Lalitpur) 2022 – 2026
Bachelor of Engineering in Software Engineering *Lalitpur, Nepal*

- Full undergraduate scholarship awarded by Pokhara University through a merit-based, national-level competitive entrance examination.
- *Relevant coursework:*
  - *Programming and theory:* Principles of Programming Languages; Data Structures and Algorithms; Analysis and Design of Algorithms; Mathematical Foundations of Computer Science (Logic, Automata).
  - *Systems:* Distributed Systems; Operating Systems; Real-Time Systems; Database Management Systems; Big Data; Computer Networks.
  - *Software engineering:* Software Testing, Verification, Validation and Quality; Software Project Management.
  - *Machine learning and mathematics:* Artificial Intelligence and Neural Networks; Probability and Queuing Theory; Numerical Methods.

Shikshodaya Technical Secondary School 2016 – 2020
Diploma in Computer Engineering *Nepal*

Research and Academic Projects

sysCrow: Log-Based Anomaly Detection and Fault Localization github.com/SatyaRajAwasth1/sysCrow
Minor project, B.E. in Software Engineering, NCIT

- Built an end-to-end system that detects anomalous behavior in system logs and points to the components and events most likely responsible for a failure.
- Implemented sequence-based anomaly detection with LSTM models, inspired by DeepLog (Du et al., ACM CCS 2017), and fault localization through unsupervised log segmentation, adapted from Dobrowolski et al. (Applied Sciences, 2024). *Python, LSTM.*

SwiftCache: A High-Performance In-Memory Caching Database
Final-year major project, B.E. in Software Engineering, NCIT

- Designed and implemented an in-memory caching database focused on low latency, high throughput, memory efficiency, and concurrent data structures.
- Implemented LRU and S3-FIFO eviction alongside a lightweight ML-based eviction policy; explored forkless snapshotting and Zstd compression for persistence, and benchmarked performance under high load. *Rust, Tokio, Zstd.*

Secure Data Migration System over TCP github.com/SatyaRajAwasth1/tcp-data-migration-system
Network Programming coursework, NCIT

- Designed a custom application-layer protocol over TCP that migrates directories by transferring only missing or modified files, detected by comparing SHA-256 hashes.
- Implemented chunked streaming, resumable transfers from the last byte received, end-to-end integrity verification, and multiplexed connections with `select()`. *C, POSIX sockets.*

Centralized Logging and Distributed Tracing System

– Built centralized logging that propagates tracing IDs across microservices, with Kafka and the ELK Stack for log ingestion and visualization. *Java, Kafka, ELK.*

Professional Experience

Niural AI December 2025 – Present
Software Engineer II *Lalitpur, Nepal*

– Architected and developed an MCP server with Accounts Payable and Accounts Receivable tools, enabling AI clients such as ChatGPT and Claude to interact with business systems and automate financial workflows through AI agents.
– Redesigned the data layer from a DynamoDB single-table architecture to a hybrid Aurora PostgreSQL–DynamoDB architecture, introducing normalized relational data models, hybrid application workflows, and stream-based synchronization between data stores.
– Designed and implemented event-driven recurring payment and invoicing workflows using AWS EventBridge, incorporating self-rescheduling execution, per-cycle state management, and approval orchestration integrating AI agents.
– Built automated production failure-detection and alerting pipelines using AWS CloudWatch, EventBridge, SQS, and Lambda, integrating incident notifications with Slack.

eSewa Ltd. March 2023 – December 2025
Software Engineer I *Lalitpur, Nepal*

– Built a real-time transaction notification system on Apache Kafka, secured with key-pair (PKI) encryption, so that more than 500K merchants could update their billing systems automatically.
– Designed and built a maker-checker administration service that lets administrators review and publish AI-recommended, personalized screens in the eSewa app, which serves 11M+ users.
– Re-engineered the SMS middleware between eSewa and the telecom operators NTC and Ncell, improving throughput, latency, and reliability for millions of daily transaction, authentication, and notification messages.
– Implemented a key-pair handshake protocol for device binding and SMS security, which also provides the basis for end-to-end encrypted communication between the mobile app and backend.
– Designed a generic QR format that lets eSewa's scanner read merchant QR codes without app changes, cutting merchant integration time by about 80%.
– Built an in-house deferred deep-linking service, a lower-cost alternative to Google's solution, to track marketing campaigns across app installs and link clicks.

Leadership, Community, and Talks

– **GitHub Campus Expert**, *GitHub Education*: selected among 70 candidates from 8,000+ applicants; delivered 10+ technical workshops and mentored at 8+ hackathons.
– **Founder and Lead**, *Nepal Tek Community (NTK)*: established and led a 10+ member core team, organizing technical events and building 6 technology-focused sub-communities.
– **Working Committee Member**, *Nepal Open Source Klub (NOSK)*: helped organize 20+ free and open source software events, including Software Freedom Day and Hacktoberfest, engaging 1,000+ participants.
– **Speaker**: Hack36 8.0 (MNNIT, India); GNOME Asia Summit 2023; GitHub Field Day 2023 and 2024.
– **Mentor**: NOSK Hackfest, MBM Ideafest, and other student-led hackathons.

Technical Skills

Languages	Java, Python, Rust, C, C++, JavaScript, SQL
Systems	Spring Boot, FastAPI, PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch, Kafka
Infrastructure	AWS (Lambda, EventBridge, SQS, ECS, S3, CloudWatch), Docker, Terraform, Linux, Git